

Electrical Installation and Workshop Technology

1. ELECTRICAL POWER SUPPLY AND DISTRIBUTION

Electrical power system

An **electrical power system** is a network of electrical components deployed to supply, transfer, and use electric power. It consists of;

- Generators: supply the power
- Transmission systems: carries power from generating centres to load centres
- Distribution system: feeds the power to industries and homes

Sources of electrical power supply

Electrical energy is produced from many different energy sources. Some of these energy sources are renewable and others are non-renewable.

Hydro

Electrical energy created from water stored in huge dams. The energy created by the water released from these dams is transformed into electricity by hydro-electric turbines and generators. The most famous source of hydroelectric power in Kenya is seven folks scheme.

Geothermal

The heat found beneath the earth surface heats water, which escapes to the surface in form of steam. This steam is used to turn generator turbines hence producing electric energy. Geothermal power plants are found in Olkaria.

Wind

Moving air (wind) is used to turn the turbines hence producing electricity. Although wind energy accounts for small percentage in Kenya, it is still an important source of electric power. Wind turbines are found in Ngong hills.

Sun

Electricity can be generated using an array of solar panels. This power can either be stored in battery banks or converted directly to AC power.

Biomass

The electric energy can be generated from landfills or rubbish dumps. This waste is collected, dried and burnt to produce heat for generating steam that turns the turbines. This type of electric power is not available in Kenya.

Non-renewable sources

Nuclear

It uses nuclear fission to generate heat, which in turn heats water to produce steam that drives the turbines. Nuclear plant is not available in Kenya.

Fossil fuels

Fossil fuels such as coal are burnt to create steam, which turns the turbines. This non-renewable source produces large amounts of electricity. This type of power supply is not available in Kenya.

Generators

Diesel/petrol or gas-powered generators are used to produce electricity. Some of the electricity that goes to the national grid comes from diesel generators.

Types of power supply system

There are two types of power supply system, which include;

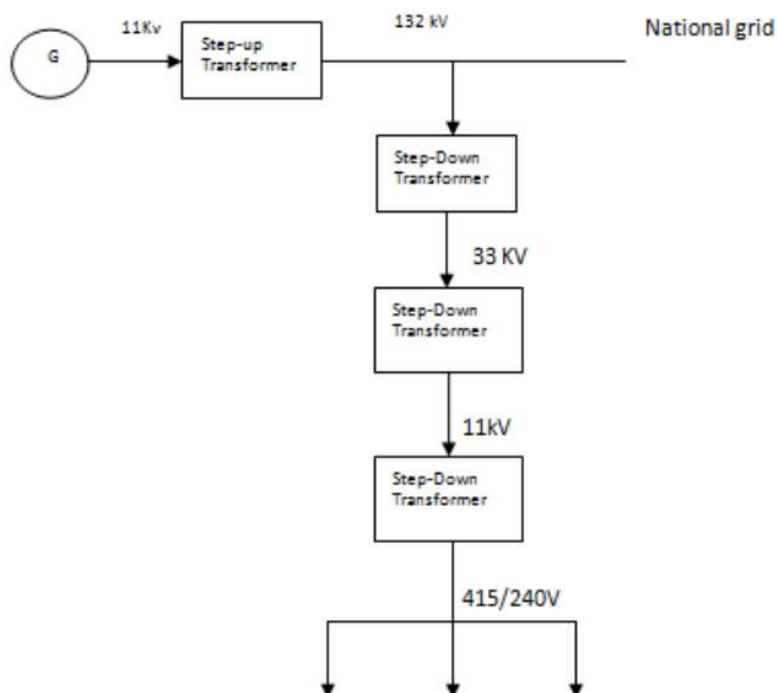
- AC power supply system
- DC power supply system

AC Power Supply system

The lines network between Generating Station (Power Station) and consumer of **electric power** can be divided into two parts.

- Transmission System
- Distribution System

The following parts of a typical power supply scheme are shown in the figure below.



Electrical power is normally generated at 11kV in a power station. This generating voltage is then stepped up to 132kV, or greater voltage. Stepping up the voltage level depends upon the distance at which power is to be transmitted. The longer the distance, higher will be the voltage level. Stepping up of voltage is to reduce the I^2R losses in **transmitting the power** (when voltage is stepped up, the current reduces by a relative amount so that the power remains constant, and hence I^2R loss also reduces). This stage is called as **primary transmission**.

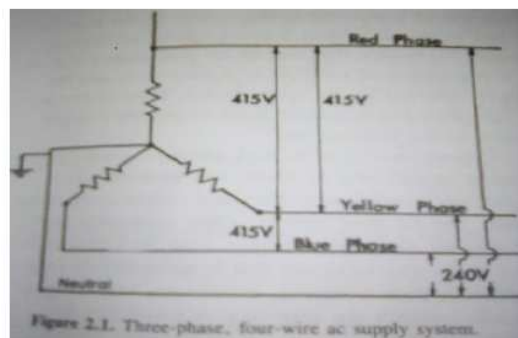
The voltage is stepped down at a receiving station to 33kV. **Secondary transmission** lines emerge from this receiving station to connect substations located near load centers (cities etc.).

The voltage is stepped down again to 11kV at a substation. Large industrial consumers can be supplied at 11kV directly from these substations. Also, feeders emerge from these substations. This stage is called as **primary distribution**.

Feeders are either overhead lines or underground cables which carry power close to the load points (end consumers) up to a couple of kilometers. Finally, the voltage is stepped down to 415/240 volts by a pole-mounted distribution transformer and delivered to the distributors. End consumers are supplied through a service mains line from distributors. The **secondary distribution** system consists of feeders, distributors and service mains.

Three phase, four-wire ac supply system

Three phase system is derived from a star connected winding of a transformer, the star point being earthed. From this point the fourth conductor, neutral is taken to form a three phase four-wire system. The voltage between any two phase conductors is 415V. The voltage between any phase conductor and neutral is 240V.



Three-phase supplies are fed to very large consumers where total load exceeds 100kW. The voltages are either 11 or 33kV. Also, three-phase is ideal for commercial and industrial premises of medium size such as schools, hospitals, hotels etc.

DC supply system

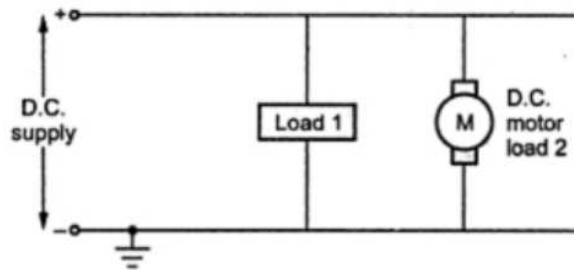
Though a.c. is extensively used everywhere, there are few applications like operation of d.c. motors, batteries, charging where d.c. supply is must. It can be obtained by using rectifiers or by d.c. generators at substations.

The d.c. systems are further classified as,

- Two wire d.c. system
- Two wire with midpoint earthed d.c. system
- Three wire d.c. system

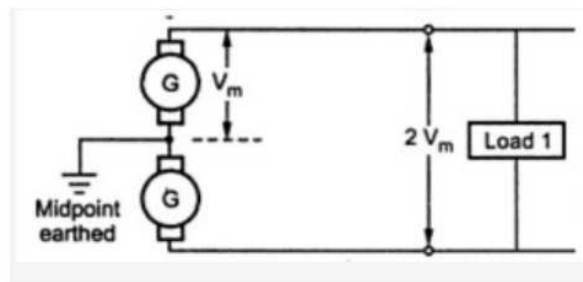
Two Wire D.C. System

It consists of two wires, one positive and other negative. The positive is outgoing while the negative is return wire. The two wire d.c. system where loads are connected in parallel is shown below.



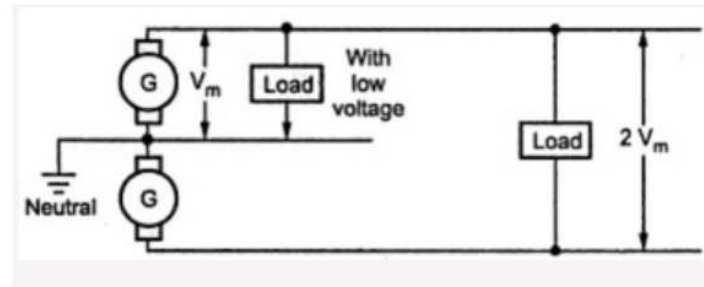
Two Wire D.C. System with Midpoint Earthed

In this system, there are two line conductors, the one having voltage of + with respect to midpoint which is earthed while the other is at potential - with respect to midpoint. Hence the total voltage available between the two lines is $2V$. No connection is taken out from the midpoint, which is earthed at the substation. The loads requiring high d.c. voltages are connected across the lines. The symbolic representation of this system is shown below



Three Wire D.C. System

This system is exactly similar to the two wire d.c. system with midpoint earthed. The only difference is, additional third wire from earth terminal of substation is available for the connections. This wire is also known as neutral wire. The scheme is shown in the figure below



When the load is balanced, no current flows through the neutral.

The loads requiring higher voltages are connected across the two outers while the loads requiring low voltages are connected between one outer and neutral as shown. The volume of copper required is higher than two wire midpoint earthed system but less than two wire one line earthed system.

2. POWER DISTRIBUTION SYSTEM

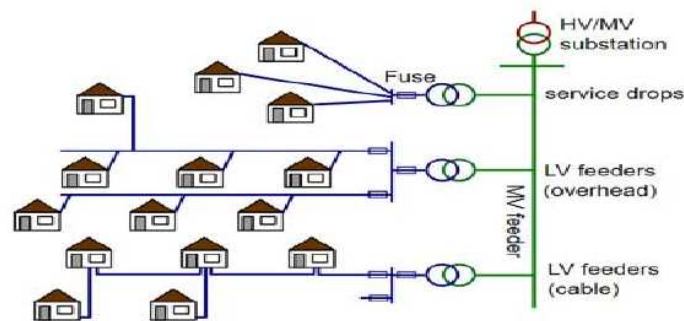
An **electric power distribution** system can be classified according to its **feeder connection schemes or topologies** as follows -

- Radial distribution system
- Parallel feeders distribution
- Ring main distribution system
- Interconnected distribution

Radial Distribution System

This system is used only when substation or **generating station** is located at the center of the consumers. In this system, different feeders radiate from a **substation or a generating station** and feed the distributors at one end.

Thus, the main **characteristic of a radial distribution system** is that the power flow is in only one direction. Single line diagram of a typical radial distribution system is as shown in the figure below



Advantages

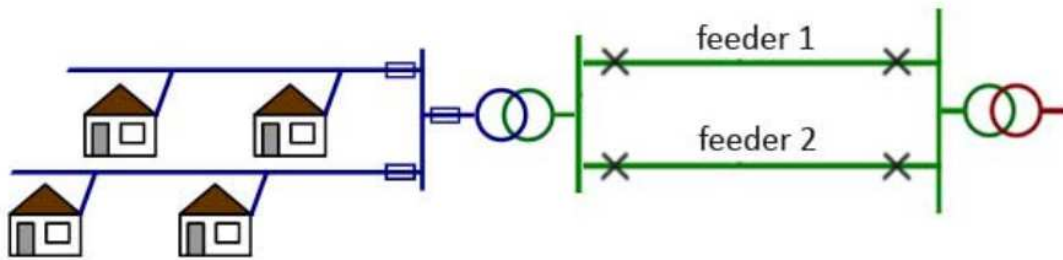
- Simplest as fed at only end.
- The initial cost is low.
- Useful when the generating is at low voltage.
- Preferred when the station is located at the centre of the load.

Disadvantages

- The end of distributor near to the substation gets heavily loaded.
- When load on the distributor changes, the consumers at the distant end of the distributor face serious voltage fluctuations.
- As consumers are dependent on single feeder and distributor, a fault on any of these two causes interruption in supply to all the consumers connected to that distributor.

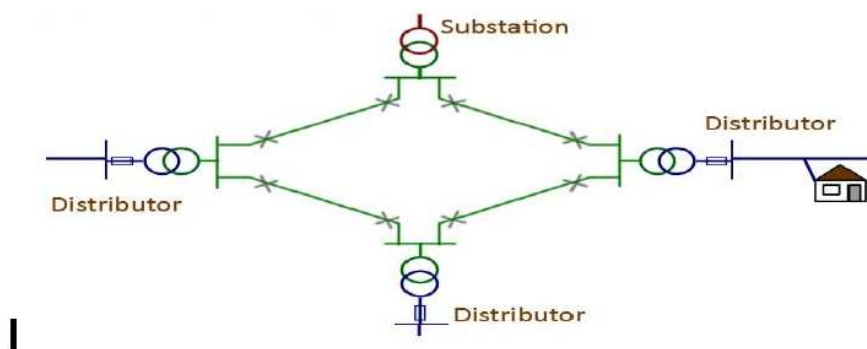
Parallel Feeders Distribution System

The above-mentioned disadvantage of a radial system can be minimized by introducing parallel feeders. The initial cost of this system is much more as the number of feeders is doubled. Such system may be used where reliability of the supply is important or for load sharing where the load is higher.



Ring Main Distribution System

A similar level of system reliability to that of the parallel feeders can be achieved by using **ring distribution system**. Here, each distribution transformer is fed with two feeders but in different paths. The feeders in this system form a loop which starts from the substation bus-bars, runs through the load area feeding distribution transformers and returns to the substation bus-bars. The following figure shows a typical single line diagram of a ring main distribution system.



Ring main distribution system is the most preferred due to its following advantages.

Advantages of the ring distribution system

- In ring power is supplied from both ends as compared to radial
- In case of a fault in the radial circuit the entire system goes off unlike in ring where by incase one end gets a fault the other end still keeps on supplying power
- Compared to the radial system, the voltage drop is less along the distribution line
- More subscribers can be installed to the system than the radial system
- Less voltage fluctuations can be seen at client's terminals. Voltage fluctuations in high loaded areas can be reduced using a tie line

Disadvantages of the ring distribution system

- Ring is very expensive n requires more materials than radial
- Radial circuit is more economical
- High maintenance cost
- It is not usable when the client is located at the center of the load

Interconnected Distribution System

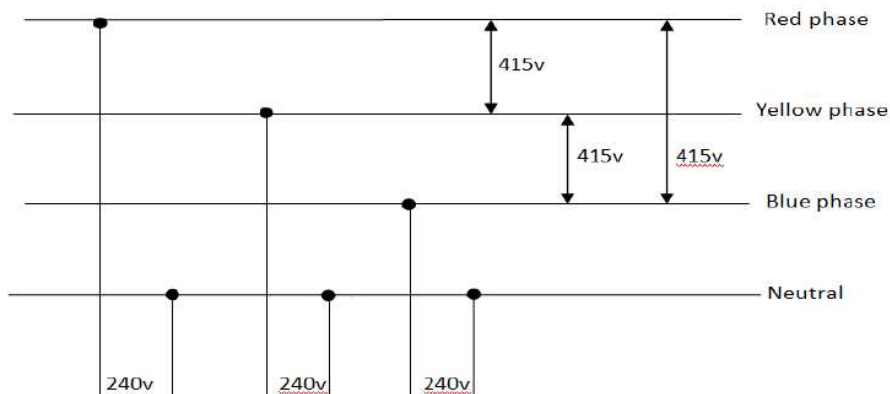
When a ring main feeder is energized by two or more substations or generating stations, it is called as an interconnected distribution system. This system ensures reliability in an event of transmission failure. Also, any area fed from one generating stations during peak load hours can be fed from the other generating station or substation for meeting power requirements from increased load.

3. POWER SUPPLY EQUIPMENT AT CONSUMER'S INTAKE POINT

The consumer can either use 400v or 240v, which is three phase and single phase respectively.

Single phase can be obtained from three phase by tapping a phase conductor and the neutral conductor. The voltage between two phase conductors is 415v.

On the other hand, the voltage between a phase conductor and neutral is 240v. Three phase voltage is derived the secondary of the transformer and converted as shown below.



After the transformer, the power conveyed to the house through house service cable. The house service cable is connected to the line conductor using mechanical connectors known as line taps.

For single phase, the service cable consists of 2 wires (phase and neutral). Three-phase consists of 4 wires (Red, Yellow, Blue and Neutral). Conductors to the premises are always insulated; in most cases Pvc-insulated.

The service cables are taken to the insulators mounted on D-iron, fixed to the walls of the house, then run to the supply intake position.



Fig D-iron

The equipment required at the supply intake position includes;

- Meter
- Supply fuse + link
- Consumer unit

I.E.E regulation

- The installation must be controlled by specified equipment, and that equipment must be accessible to the consumer and shall be near to the supply-intake cutout.

In this case, the switchgear should be installed at the supply intake position. The switchgear must be able to;

- i. Isolate the installation from the supply
- ii. Protect the installation against excess current which arise in case of short circuit
- iii. Cut off the current if an earth fault occurs when live conductor touches the metal.

SEQUENCE OF SUPPLY CONTROL EQUIPMENT

- The first item in the supply supply-control equipment is the fuse and neutral link. The fuse allows maximum current to flow when total load is connected to the supply. In most cases, the domestic installation requires a fuse rated 60A. These fuse can be rewirable or cartridge type. The neutral link is a solid link of flat tinned copper is used where the neutral side of the supply is effectively earthed.
- The second item in the sequence is the meter. The purpose of the meter is to record the amount of electrical energy of current-using items connected to the electrical installation. It records the product of volts (V), current (I), and time (t). The standard unit of meter is Kilowatt-hour or kWh. The meter is sealed against tampering and unauthorized entry.
- From the meter, the installation cables are taken to the switch or switchfuse or consumer unit. The purpose of the switch is to isolate the complete installation from the supply when switch is operated.

NB

The switchgear must be fitted as close as possible to the meter and the main cut cutouts, so the cables from the meter must be short as possible.

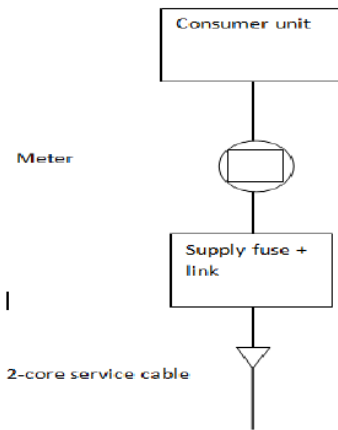


Fig (a)

Fig (a) – sequence of supply control in single phase

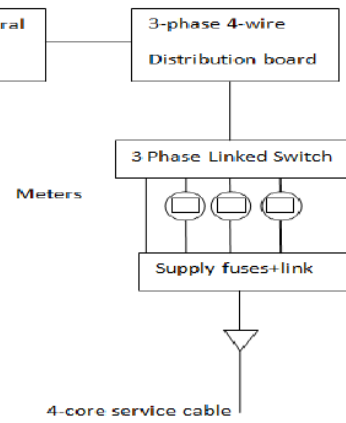


Fig (b)

Fig (b)-sequence of supply control in three phase

4. CONDUCTORS AND CABLES

Conductor is a material that conducts electricity

Cable is a conductor that consists of insulation and mechanical protection

Cables form necessary connection between power supply and the apparatus which uses it. It should conduct electricity, cheaply and safely.

Conductor materials

Copper and aluminium are used as conductors in power and lighting cables.

i) Copper conductors

They are annealed or hard drawn. Annealed copper are soft and pliable and are most suitable for indoor and outdoor wire and cables laid in fixed position.

Hard drawn copper cables have very high tensile strength and are used as overhead wires mainly in bare form.

ii) Aluminium conductors

They are made of standard size but are used only large sizes.

IEE regulation: All cable conductors of cross-section area of 10mm^2 or smaller shall be of copper.

Insulation

It is the protective layer that confines electricity to the conductor. It must have high resistance. The insulating materials used are:

- Polyvinylchloride (p.v.c)
- Impregnated paper
- Mineral insulation
- Elastomers

Mechanical protection

The cables must be protected against mechanical damage. We have **unarmoured** cables and **armoured** cables.

Flexible cord and cables

They are made of fine gauge wires and are more flexible than ordinary wiring cables. They are used for such purposes as from ceiling rose to lampholder or from socket outlet to portable apparatus.

IEE regulations concerning flexible cord s and cable

- The type of insulation and protective covering shall be such as to minimize the risk of damage to installed cable
- General purpose p.v.c cable shall not be installed where temperature is consistently below 0 degrees
- Flexible cords and cables exposed to risk of contact with water shall be sheathed with rubber or p.v.c
- It is recommended that heat resisting flexible cords be used between ceiling rose and lamp holders in pendant fittings where tungsten filament are fitted
- Flexible cords and cables shall not normally be used as fixed wiring
- Exposed length or flexible cord shall be as short as possible

General IEE regulations concerning cables

- Single core steel armored cable shall not be used for alternating current
- Where cables are connected in parallel they must be of the same type, size and length to ensure proper division of current
- All conductors and cables shall be adequately protected against mechanical damage
- All cables in an installation shall be supported so that there is no mechanical strain on the cable termination
- Where cable pass through walls or partitions the surrounding holes must be made good with cement or other fire resisting material to prevent spread of fire
- Exposed lengths of flexible cables or flexible cord shall be as short as possible

Factors affecting current rating of the cables

- Ambient temperature
- Class of excess-current protection
- Grouping
- Disposition
- Type of sheath

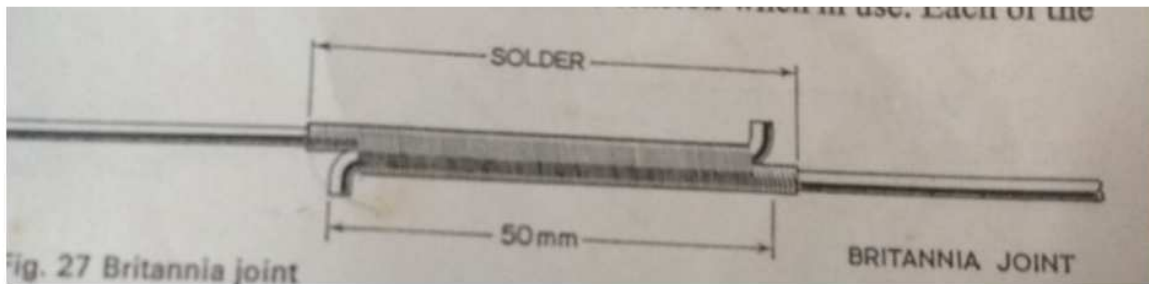
Joints and jointing

The IEE regulation requires that cable conductors and bare conductors shall be electrically and mechanically sound.

Types of joints

i) Britannia joint

This joint is used for single overhead copper wires which are in considerable tension when in use.

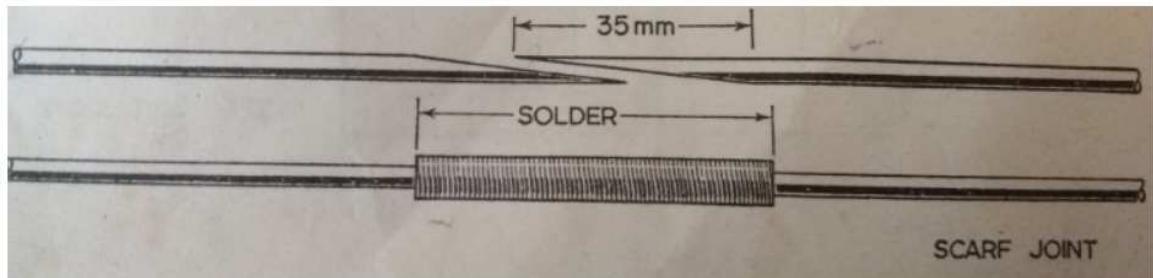


Procedure

- Each of the two wires to be joined is thoroughly cleaned and tinned for a distance of 75mm.
- About 6mm of the end is bent over at right-angles and forms as stop to prevent joint from pulling out.
- The two wires are then brought together over a distance of 50mm with the stops in opposite directions and held in a vice with pliers.
- They are then bound together with suitable binding wire.
- The binding is carried past the stop at each end for four or five turns.
- The whole joint is then soldered
- No insulation is required

ii) Scarf joint

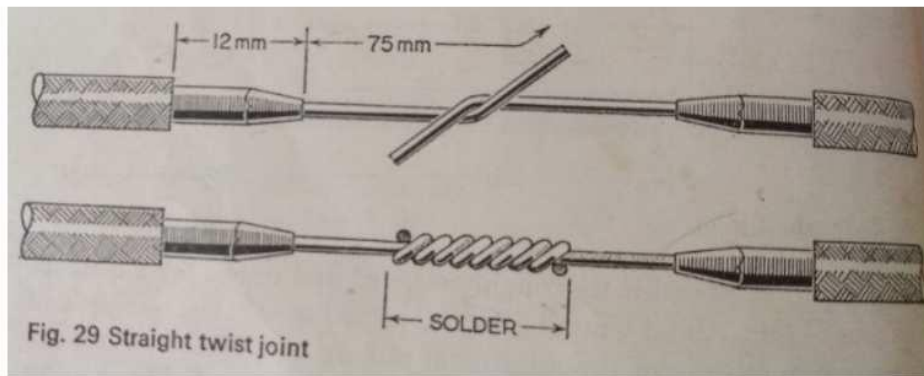
This joint is suitable for wires which are not required to stand tensile stress and when joint to be as small as possible.



Procedure

- Each of the two ends is carefully chamfered by filing over a distance of about 35mm
- The ends are cleaned, tinned, and fitted together so as to have the same diameter as an uncut wire.
- The two wires are held tightly together and wound with binding wire over distance of 60mm.
- The whole joint is then soldered.

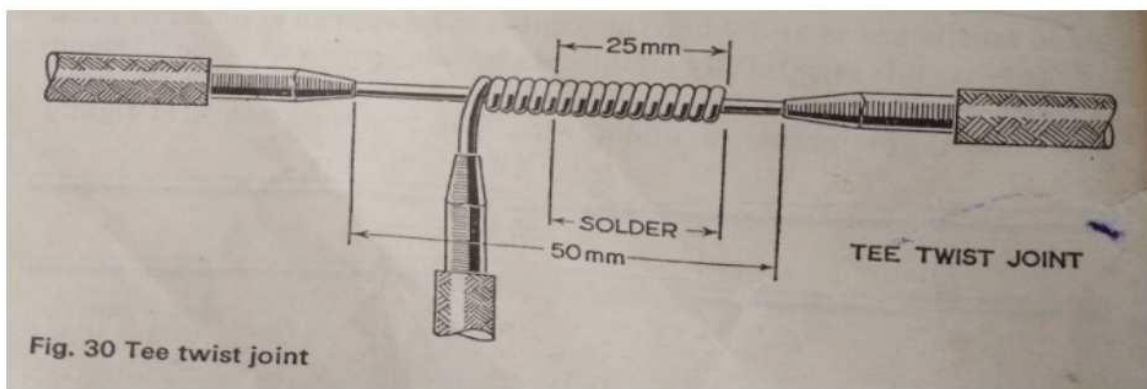
iii) Straight twist joint



Procedure

- Each end is prepared, cleaned and tinned over a distance of 75mm.
- The wires are then laid together about 50mm from the ends and twisted tightly round each other in opposite directions each turn of wire fitting closely up to the next.
- The joint is then soldered from end to end.

Tee twist joint

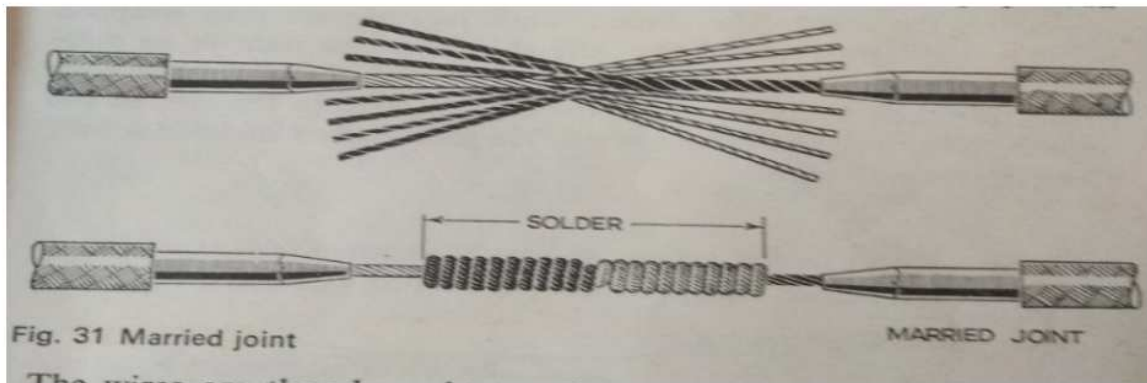


Procedure

- The through wire is bared, cleaned and tinned for a distance of 50mm. The wire is not cut
- The tee wire is similarly prepared, and 50mm of its length is tightly bound round the through wire from, say left to right.
- In soldering the joint it is common practice to leave the first four turns unsoldered to allow flexibility when insulating the joint.

iv) Married joint or stranded cable

When properly made this is very strong and satisfactory joint.

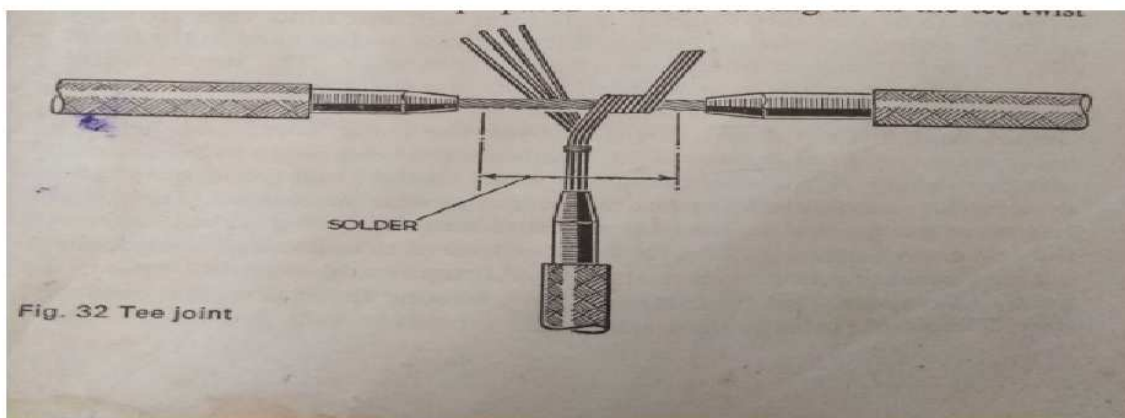


Procedure

- In preparation of the ends, 75mm of wire is stripped of insulation.
- The strands of each are firmly twisted together in direction of the lay for 25 mm, leaving 50mm splayed out.
- The wires are brought together, end to end, each strand lying between two strands of the opposite cable.
- The strands of the right hand cable are held down along the left hand cable, while the strands of the right hand cable close together half a turn at a time.
- When strands are completely bound round they are tightened up with pliers, and the other side is similarly completed, the binding being in the opposite direction.
- The joint is then soldered

v) Tee joint on stranded cable

The figure below shows a tee joint on the 4.0 mm² cable



Procedure

- The through wire is prepared without cutting as in the tee twist joint for a distance of 75mm.
- The tee cable is then cleaned for 75mm, 25mm of which is twisted in the direction of the lay, and secured with two or three turns of binding wire to prevent it from untwisting.
- The tee strands are divided as shown in the figure.
- The tee wire is now brought to the centre of the through wire with three strands on one side and four strands on the other.
- The three strands are together are then wrapped carefully and tightly round the through wire to the right in an anticlockwise direction, and finally tightened with pliers.
- The four strands are then wrapped round the through wire to the left, in the opposite direction and tightened up.
- The joint is then soldered solidly along the through wire.

ELECTRICAL WIRING SYSTEM

A wiring system is an assembly of parts used in formation of one or more electrical circuit.

It consists of a conductor together with its insulation, which serves as a protection against mechanical damage.

Factors to consider when choosing a wiring system

- **Type of building;** whether the installation is for permanent or temporary building or for extension to an existing building.
- **Installation conditions;** whether the installation is likely to be subjected to mechanical damage, moisture, fumes , gas, inflammable or explosive dusts, etc
- **Flexibility;** whether the wiring system will allow alteration or modification in the future
- **Appearance ;** whether the installation should be hidden or visible
- **Durability;** whether the installation will last for a time of the life of a building or not

- **Cost;** whether there are resources to carry out the installation in the building
- **Safety;** depends on the installation conditions type of the supply and the earthing arrangements available.

Types of wiring systems

a) SHEATED WIRING SYSTEM

This includes all wiring which use sheath as its mechanical protection. There several types depending on the material used for sheating, they are;

i) Tough Rubber Sheath (TRS)

It is termed as all-insulated system because the sheath is of an insulating material. The insulation used is tough rubber and the sheath is tough rubber. The cable is available in single core, flat twin, and three core

Advantages

- Cheap
- Flexible
- Easy to install

Disadvantages

- Rubber is affected by oil
- Tend to harden at high temperatures

ii) Polyvinylchloride (PVC)

It is an all-insulated wiring system where both the insulation and sheath are pvc. Available in single core, flat twin with earth and 3 core with earth.

Advantages

- Less affected by oil

- Has less insulation resistance

Disadvantage

- Soften when warmed therefore excluded in a number of installation applications

iii) Polychloroprene sheathed (p.c.p)

The cable consists of conductor insulated with vulcanized and sheathed with p.c.p, which is a tough rubber like substance. This plastic like material is suitable for farm wiring where steam, ammonia fumes, lactic acid and milk fats, sulphur fumes, direct sunlight and heat is likely to be encountered.

iv) Lead alloy sheathed (LAS)

This system consists of conductors which are insulated with vulcanized rubber, taped and sheathed with lead alloy. The lead sheath serves as a circuit protective conductor and protection against mechanical damage.

Disadvantages

- Lead sheath is very soft and can be damaged by hard objects
- LAS cables are prone to corrosion when they come in contact with damp environment

v) Mineral insulated metal sheathed (MIMS)

The cables consist of aluminum conductor contained in copper all aluminum sheath. The insulant is compressed mineral magnesium oxide. They are resistant to water, oil and immune to condensation. They do not require further protection even against high temperatures and fire. The conductor sheath and insulant are inorganic hence the cables are ageless. During installation the ends of the cable must be sealed off against the ingress of moisture by special installations.

Advantages

- The cable has good heat resisting properties
- Has a higher current rating than PVC, TRS
- Offer excellent circuit protective conductor since sheath is copper or aluminum

Application: for industrial and hazardous areas

vi) House service overhead system (HSOS)

In this system, the conductors are insulated with vulcanized rubber. The conductors of the cables are calico tapped, then coated with weather resisting compound. This type of wiring system is used for consumer's distribution wiring.

b) CONDUIT WIRING SYSTEM

A conduit is a tube used to protect and route electrical wiring in a building or a structure. The conduits can be metallic or non-metallic

i) Metallic conduits

Steel conduits

They are available in two classes; class A and class B.

Class A conduits are not strong enough to withstand threading. They are connected to various electrical accessories using a lug grip. The conduit is held securely by tightening screws in the lug. Class A are used in situations that are not damp and where the wires do not require high degree of protection against mechanical damage.

Class B conduits are threaded at the ends. They are more strong and restricted to gas tight and explosion proof installation. Class B conduits include heavy gauge welded conduits and solid drawn conduits. The finishing's used in these conduits are;

- Black enamel for internal use in dry situations
- Silver grey finish for use in dry situation where conduit is required to match decoration
- Hot galvanized or stainless steel for external use where the conduit will be subjected to dampness or condensation

Advantages of steel conduits:

- Provide protection against mechanical damage
- Provide earth continuity conductor (ECC)
- It is durable
- It can be easily extended
- It is fire resistant

Disadvantages

- It is more expensive
- It is difficult to conceal
- It is prone to corrosion

Copper conduit

They resist corrosion and is an excellent earth continuity conductor (ECC). They are expensive but have extremely long life.

Aluminum

It is inferior to steel but has found applications due to its light weight. It is an excellent Earth Continuity Conductor (ECC). It may require further protection since it is not mechanically strong.

Flexible conduits

It is used for final connection to the machinery e.g electric motors where vibrations and the need to adjust to the position of the equipment make a rigid conduit connection is unsatisfactory.

ii) Plastic (PVC) conduits

Available in two types:

Light gauge: this is a flexible type which has a round cross sectional area and is supplied in 4 meters length. It can be bent without tools, but bending spring or sand is inserted to prevent it from collapsing at bends.

Heavy gauge: this type is bent carefully by use of a bending spring. It is supplied in several lengths. Bending is done by inserting a bending spring whose diameter is slightly less than the internal diameter of the conduit and you bend it by putting across the knee.

Advantages

- Not expensive
- Does not pose risk of earth leakage thus no shock risks
- It is resistant to corrosion
- Easy to install
- It simplifies the routing of conduit where there are obstacles

Disadvantages

- It expands and contracts as temperature changes causing change in internal cross sectional area
- Its insulation resistances and mechanical strength are reduced at temperature of 120 degrees and above

NB: in order to comply with IEE regulations the PVC should have an earth wire run inside the conduit

IEE regulations regarding electrical conduits

- The conduit installation must be completed before cables are drawn in
- The space factor of 40% is allowed in the conduit system

- The end of the conduit must be filed or reamed to prevent damage to the cable
- Conduit installed in damp conditions must have water resistant finish e.g galvanized.
- Extra low and high voltage cables must not be run in the same conduit.
- Metal conduit should be kept separate from gas and water serving conduits.
- All metal conduits must be effectively earthed and all joints must be mechanically and electrically sound and continuous.
- Drainage points should be provided at the lowest point in a conduit installation and in conditions where condensed moisture is likely to collect. Drainage point should not be made in gas tight system.
- Cable installed in explosive atmosphere must be enclosed in solid conduit unless they are metal sheathed.

c) TRUNKING WIRING SYSTEM

Trunking is available in square or rectangular sections ranging from 1.5 inches by 1.5 inches to 12 inch by 6 inch in cross sectional and length of 2 meters. Trunking is very rigid and cannot be bent or set around obstacles.

Forms of trunking

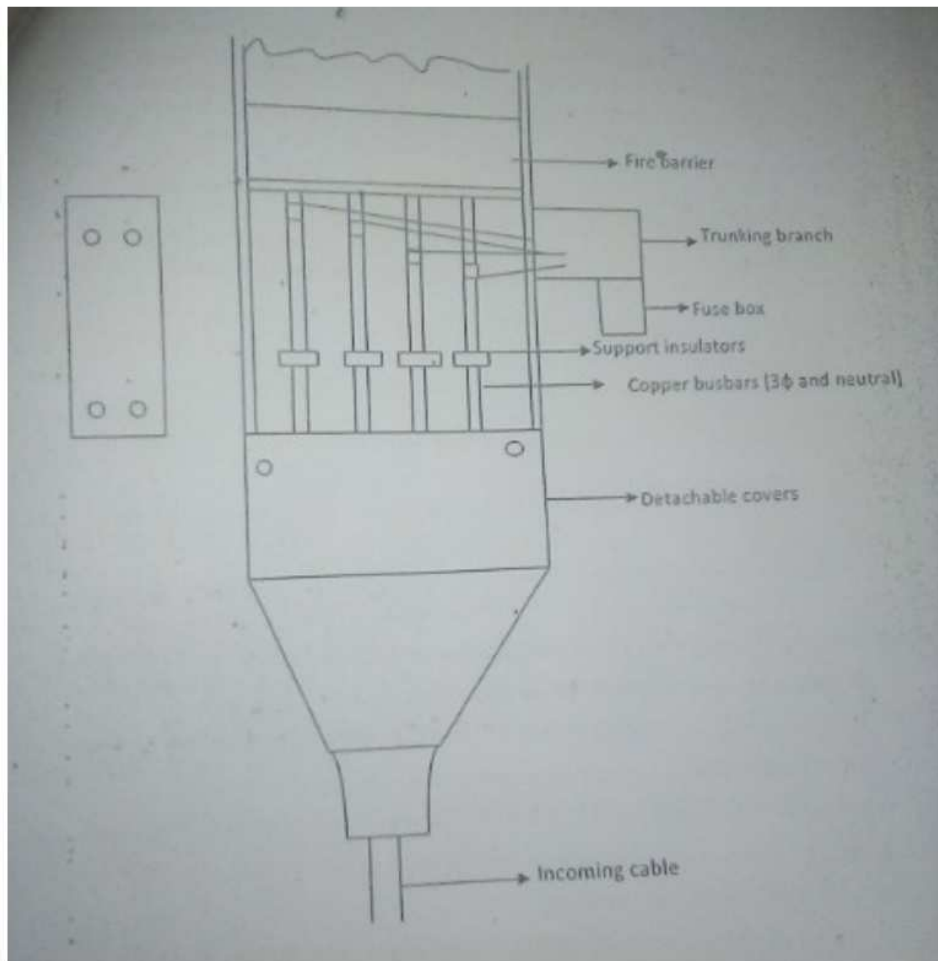
- Busbar trunking
- Flush trunking
- Multi-compartment
- Skirt trunking
- Lighting
- Cable tap trunking
- Pvc trunking

a) Bus bar trunking system

It is a wiring system which consists of copper busbars supported on insulators and enclosed in steel trunking which is convenient for use where large currents have to be handled. Application of this system is:

Vertical rising mains: it is often employed in a multi-storied building to carry supply to each floor. The incoming supply is connected to the bottom of the rising main and suitable tap off point provided at each floor level.

A common type of trunking used for this purpose consists of 300A copper busbars. Since in the event of fire hot gases and flames tend to travel up the trunking, fire proof barriers are fitted at each floor level. A typical fire barrier consists of 2-3 inch thick layer of glass wool supported by hard asbestos board sheet and completely fills the trunking.



Fig(1) vertical rising mains

b) Flush trunking

It fits flush with the walls. It requires a lot of builders work to install the system.

c) Multi-compartment

It is provided with segregated compartments so that cables carrying different voltages can be accommodated in the same trunking unit.

d) Skirt trunking

It is designed to take the normal place of the room skirting. It carries power, telephone and lighting cables in its various compartments.

e) Lighting trunking

It is designed for use where long rows of continuous lighting are required. The steel enclosure can carry fluorescent and tungsten fittings, control gear and supply cables.

f) PVC trunking

Used for lighting circuits mostly in extensions of buildings, bell circuits and for telephone cables.

g) Cable-tap trunking

It does not carry copper busbars but insulated supports which can accommodate VRI and PVC cable from which supplies to machines and lighting circuits are tapped through fused tap off boxes.

Advantages of trunking

- It is much lighter than conduits of the same capacity
- Has fewer fixings
- Wiring is easier and quicker as cables are laid instead of being drawn in
- Installation time is reduced
- Multi-compartment is available where segregation of services is needed

Disadvantage

- The initial cost of installation is high

IEE regulations regarding trunking

- Where cables are having different ratings are installed in the same enclosure, all cables shall be deemed to have lowest temperature rating
- Busbar trunking must not be installed in conditions where inflammable vapours are present
- Fire barriers must be installed inside the trunking when it passes through the floors, walls and partitions. Floors, walls and partitions must be made of incombustible material after the trunking has been installed
- Allowance should be made for expansion in long sections e.g by fitting copper braiding between sections
- All busbar trunking should be marked danger and voltage stated and lids must be securely fitted
- Spacing factor must not exceed 45%

d) UNDER FLOOR DUCTS WIRING SYSTEM

The ducts are made of chemically inert non metallic fibrous material impregnated with a preservative bitumen compound. The ducts are installed in the solid concrete floors during the erection of the building. Various ducts are arranged to feed ceiling points for the floor below and for the floor above. Junction boxes are provided with cover plates fixed flush with floor finish and since the ducting is laid out between junction box, it is always possible to locate the runs.

Advantages

- The major part of the accommodation for the wiring can be conveniently installed at an early stage during the construction of the building even though the exact positions of the outlets may not be known at that time
- A concealed wiring system is provided without the need for excessive long chases in the concrete floors
- It is relatively easy to alter the positions of outlets if necessary e.g in case of change of tenancy

IEE regulations regarding under- floor ducts

- The ducting system shall be sufficiently protected to prevent damage by the intended use of the floor
- The ducting shall not be run in such a position that is not liable to damage by contact with the floor or the ceiling or their fixings
- Entries into the ducts must be protected from ingress of water. All covers should be securely fixed on completion
- A maximum space factor of 35% must be ensured
- Where ducts pass through walls or partitions, the hole should be plugged of fire resisting materials to the thickness of the building materials to avoid spread of fire

SPECIAL INSTALLATIONS

Special installations deals with wiring that have been developed to meet the requirements of specific installations. They include

a) Bare conductors

They are not insulated and are installed for the following purposes;

- Earthing connections
- Protected rising main and busbar system
- The conductors of extra low voltage systems
- As collector wires for travelling cranes and trolleys

Bare conductors passing through the walls, floors, partitions or ceilings must be protected by enclosing them in a non absorbent incombustible insulating material

b) Catenary systems

It uses high tensile steel wire pulled taut between buildings. The cable is then run along catenary wire being supported the same way. This system is also known as grid suspension system and consists of central galvanized steel, high tensile stranded wire surrounded by a number of pvc insulated conductors and the whole is enclosed with pvc tape and pvc sheathing. Special connecting boxes are used at tees and right angle turns.

Advantages

- The system is completely waterproof
- Weight imposed on the building is reduced

- Installation can be left until building is completed
- Has reduced installation costs

Applications

- In industrial buildings and similar installations where other type of wiring systems would be either difficult or expensive to install
- For overhead street lighting and factories instead of an overhead busbar system where the amount of power required for machines is low

c) Earth concentric wiring

This consists of one core tinned copper made up with an outer sheath of tinned copper. The functions of outer sheath is to act both as return conductor for circuit current and as a mechanical protection

IEE regulation

Earth concentric system may only be used if it is supplied by a transformer and has no metallic connection with the public supply system.

d) Cleated wiring system

It consists of VRI cables supported by plastic or porcelain cleats.

IEE regulation

- The conductors in a cleated system must not touch one another or adjacent surfaces e.g ceilings or walls
- The conductors must be in view throughout their length and must be supplied with mechanical protection when they pass through the walls.
- Conductors must be terminated at a non inflammable surface e.g. hardwood or block

NB: this system is very cheap to install but it is only used in conditions where it is not liable to damage from moisture or mechanical damage

e) Overhead system

It is not an actual wiring system but an overhead cable which provide distribution of electric energy between points of use and between buildings where there is some considerable distance between the main control of supply and point of use e.g. a farmhouse and water pump mile away.

f) Temporary installations

A temporary installation is an installation with an expected period of service of three months. Any installation beyond this point should be completely overhauled.

IEE regulation

- Every installation on a construction site, temporary installation other than in private dwelling house shall be under the charge of competent person who shall be fully responsible for the safety and use of the installation and for any alteration or extension
- The installation shall be inspected and tested at intervals of three months or less, as necessitated by the nature of installation

Important electrical wiring accessories

Electrical accessory is a basic part used in wiring either for protection or for the control of the electrical circuits.

Accessories are classified into the following:

- Controlling accessories
- Holding accessories
- Safety accessories
- Outlet accessories
- General accessories

i. Switch

A switch is used to make or break the electric circuit. The switches may be one-way switch and two-way switch. The one-way switch is used to control single circuit or lamp, where as the two-way switch is used to divert the flow of current to either of two directions. The two-way switch can also be used to control one lamp from two different places (As in case of stair case wiring). The different types of switches are available in the market, depending upon the requirement; few important are detailed below

a) Surface Switch: The switches are mounted on wooden board, fixed over the surface of the wall. These switches are also known as “Tumbler switches”.

b) Flush/Piano type switch: The switches are used where good appearance is required. The switches are fixed in flush with the wall and do not project out. These switches are also known as “Piano Type Switches”.

c) Bed switch: As the name indicates, it is used to switch “ON or OFF” the light from the

place, other than switch-board or from near the bed, while going to sleep or getting up. This switch is connected through flexible wire.

d) Rotary switch: This switch is used to control different lamps from one places one by one or as selector switch, to select different voltage tapping of transformer in voltage stabilizer.

e) Push Button switch: These switches are used to control the electric bell and indicating lamps etc. When the push button is pressed, the circuit is completed and the bell or lamp is switched on. The supply to bell or lamp is switched off as the push button is released.

f) Pull switch: These switches are operated with a single pull of the cord, for the on and off position. These are also used in bedroom and bathroom. These switches are fixed near the ceiling and hence these are also known as “Ceiling Switches”.

g) Intermediate switch: This switch has four terminals and four different connection positions. The main function of this switch is to control a lamp from three or more different places, along with ordinary two-way switch. Generally this switch is used in double stair case wiring or corridor wiring. This switch is also known as four-way switch.

h) Lamp Holder: As the name indicates, a lamp holder is used to hold the lamp, and connect it electrically to supply terminals, required for lighting purposes. Some important types of lamp holders are detailed below:

a) Batten Lamp Holder: These lamp holders may be of brass or bakelite, with brass plunger. The holders are fixed on either on round block or wooden board with the help of wooden screws.

b) Pendant Lamp Holder: This lamp holder is used to hang the lamp from ceiling rose, with flexible wire. Some time these holders are provided with lampshades, to divert the upward light to down ward. These may be of brass or bakelite, with brass plunger.

c) Angle Lamp Holder: The angle holder is used to focus the light at an angle and is fixed directly on wall or round block with the help of gutties and wooden screws. These may be of brass or bakelite with brass plunger.

d) Swivel Lamp Holder: These lamp holders are used for lighting of shop windows, show case etc. It consists of ball and socket joint fitted between back plate and lamp holder, for the purpose to move the light to a wide angle.

e) Bracket Lamp Holder: These lamp holders are used to focus the light on the floor or at some angle, slightly away from the walls. Light shades can also be used for diverting all light

on floor. Such lamps are provided with such fittings which make them water tight so that these can be used out side the houses or for street lighting. The bracket may be of wood, aluminium and brass. The lamp holder is simply a pendant holder made of brass or bakelite.

iii. Fluorescent Lamp (Tube) Holder

These are used to hold the fluorescent tube and pin type holders are generally used

iv. Socket

The sockets have insulated base with moulded base having three terminal sleeves. The two terminal sleeves having same cross-section are used to connect phase and neural wire where as the third having greater cross-section is used to connect with earth wire. These are made for 5 amps and 15 amps load. Two-pin sockets are also available, in which only phase and neutral wires are connected. These are available in 5-amp capacity. These may be surface type or piano-types. The surface type accessories are obsolete now days.

vi. Plug

The plugs are also having moulded three pins of brass or any electrically conducting material. These are also made of 5 amp and 15-amp rating. These are used for taking power from socket.

vii. Wiring clips: used to fix the wires to surface.

ELECTRICAL WIRING TOOLS & EQUIPMENT

i. Wire Tester



Essential diagnostics tools, [wire testers](#) indicate whether the power is off or on, which makes them essential for safety reasons. Moreover, they also determine the strength of electricity, with advanced models offering digital readouts. The best wire tester is a multimeter that provides both voltage and continuity testing.

ii. Channel Lock Pliers



A versatile hand tool and useful electrical tool, [channel lock pliers](#) have several tasks to perform. They are useful when tightening connectors in electrical boxes, removing knockouts from the boxes, and adjusting expansion-type ceiling fan boxes.

iii. Screwdriver



Every handyman should have a set of [screwdrivers](#). A small cordless screwdriver serves as a perfect addition. It comes handy when ~~driving in~~ screws in electrical boxes, electrical fixtures, or when screwing the outside plates onto the wall or ceiling. An adjustable screwdriver suits most applications. For electrical work, a wire-bending screwdriver becomes handy. Many also feature insulation for safe use.

iv. Hammer



A [hammer](#) is a multifunctional tool also necessary for electrical work. It helps secure boxes that come with nail-on brackets and drives in straps in case of wiring. A basic one with a claw and a flat head is sufficient.

v. Wire Strippers



Wire strippers remove the electrical insulation off electrical wires without damaging the interior wire. Manual models require some care during use, while automatic ones allow even novice electricians or DIY fans to remove insulation quickly, although the latter works with a certain size range of wires. Representing the most advanced models, laser wire strippers burn the insulation around the wire.

vi. Conduit Bender



A **conduit bender** becomes extremely useful when replacing an old electrical setup, wiring a new space or home, or simply rearranging the wires to suit the furniture placement. Good conduit benders perform common bends, such as saddle, back-to-back, offset, and stub-up, and also feature scales for accuracy.

vii. Power Drill



A handheld [power drill](#) constitutes one of the most common hand power tools that electricians require. Generally, 18-volt models provide sufficient power to drill through any kind of material, while 12-volt pocket models are handy for their portability. Rechargeable lithium-ion batteries make it possible to use it away wirelessly, so electricians can get to even the hard-to-reach corners

viii. Utility Knife



A [utility knife](#) cuts through many materials and thus serves as an essential for any electrical work. Portable and practical, they quickly fold into a more compact form, and many feature a belt clip for easy carrying. This becomes especially useful when climbing the ladder, as it leaves the hands free.

ix. Tape Measure



A basic part of any toolkit, a [tape measure](#) helps when it comes to fitting new electrical fixtures and centering lighting fixture boxes. measuring wall height and depth as well as wire length represent just some of the common electrical works. When fitting new fixtures, a level perfectly complements a tape measure.

EXCESS CURRENT PROTECTION

Excess current protection is one the requirements of IEE regulations which states that: every consumer installation shall be adequately controlled by switchgear readily accessible to the consumer which shall incorporate means of excess current protection.

An excess current is a current which flows in a conductor and has value which is in excess of the maximum permitted rated current of the conductor.

Excess current results from:

- Sustained overloading
- Heavy sudden overloading
- Short-circuits

Sustained overloading occurs when extra power is taken from the supply system. Overloading results in heating up of the cables which may lead to the fire outbreak.

Heavy sudden overloading occurs when overload current flows for a short time and is not serious as compared to sustained overloading.

A **short circuit** is the direct contact between live conductor and neutral conductor or earthed metalwork.

The circuit can be protected using a fuse or circuit breaker

i. Fuse

It is a device for opening a circuit by means of a conductor designed to melt when an excessive current flows along it. The following are terms used in connection of the fuses:

Fuse element: this is the part of fuse that is designed to melt when excess current flows through it.

Current rating: the maximum current which the fuse will carry for an indefinite period of time without deterioration of fuse element.

Fusing current: this is the minimum current that will cause the fuse element to heat up and melt or blow.

Fusing factor: this is the ratio of the fusing current to the current rating.

$$\text{Fusing factor (F)} = \frac{\text{Fusing Current}}{\text{Current rating}}$$

Types of Fuses

- a) Rewirable
- b) Cartridge
- c) HRC (High-rupturing capacity)

Rewirable fuse is a simple device that consists of short length of wire. The current at which the wire melts depend on the length and cross sectional area of the wire. Though it is cheap rewirable fuses has a number of limitations.

Cartridge fuse has a current rating that is accurately known. It is more expensive to replace than rewirable fuse.

HRC fuse has short time characteristics that enables it to take care of short circuit conditions is the protection of motor circuits.

ii. Circuit breaker

It is a mechanical device for making and breaking a circuit, both under normal and abnormal conditions such as short-circuit.

The circuit breaker differs from a switch. The switch is capable of making and breaking a current not greatly in excess of its normal rated current.

The circuit breaker is capable of disconnecting automatically a faulty circuit, even in short circuit conditions. Also the circuit breaker interrupts the circuit fault without damage to itself.

Circuit breakers provide more accurate degree of excess-current protection as compared to fuses.

SOLAR POWER SYSTEMS

We can change sunlight directly to electricity using solar cells. Every day, light hits your roof's solar panels with photons (particles of sunlight). The solar panel converts those photons into electrons of direct current ("DC") electricity. The electrons flow out of the solar panel and into an inverter and other electrical safety devices. The inverter converts that "DC" power (commonly used in batteries) into alternating current or "AC" power.

Solar accessories

Accessories are things needed to install solar panels so that they can set up and operate their panels efficiently.

1. Solar Panel Batteries

They are designed to store the energy that solar panels generate during the day. The batteries respond to changes in the amount of power produced by solar panels to help maintain steady wattage outputs.

2. Solar Inverters

They are an essential part of a solar panel set-up because they convert the DC energy of the sun to electrical power.

3. Solar panel Frames

They are used to mount the solar panel on the roof of a building.

4. Solar Charge Controller

During bright, sunny days, a solar panel series may produce more energy than its battery and immediate output can use. To protect a solar panel set-up from the damage that can result from excessive voltage generation, you will need a solar charge controller.

Grid connected solar power system

A grid-connected photovoltaic power system or grid-connected PV power system is a solar PV system that is connected to the utility grid. A grid-connected PV system consists of solar panels, one or several inverters, a power conditioning unit and grid connection equipment. They range from small residential and commercial rooftop systems to large utility-scale solar power stations. Unlike stand-alone power systems, a grid-connected system rarely includes an integrated battery, as they are still very expensive. When conditions are right, the grid-connected PV system supplies the excess power, beyond consumption by the connected load, to the utility grid.